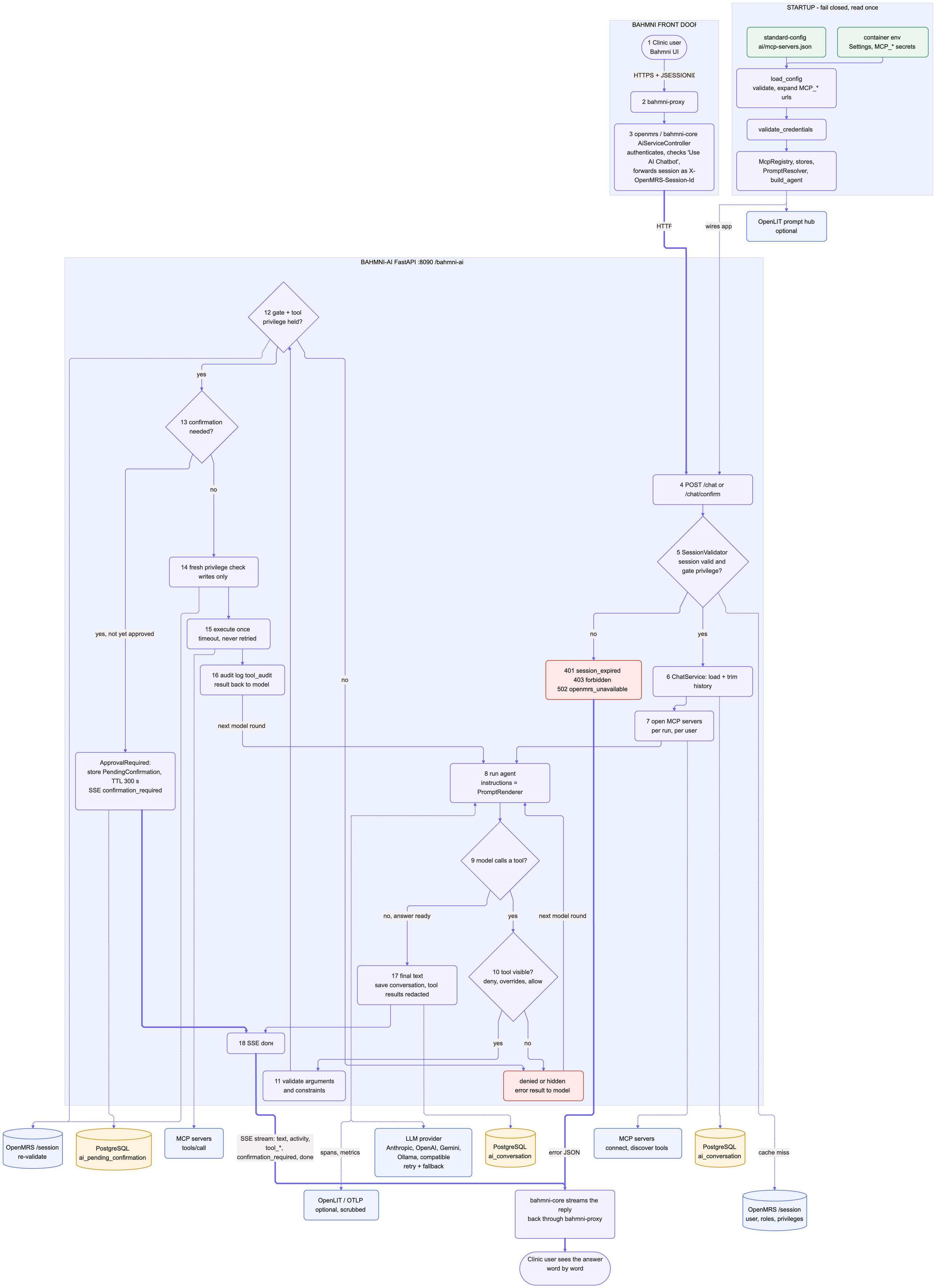

BAHMNI FRONT DOOR
1 Clinic user Bahmni UI
HTTPS + JSESSIONID
2 bahmni-proxy
3 openmrs / bahmni-core AiServiceController authenticates, checks 'Use AI Chatbot', forwards session as X-OpenMRS-Session-Id
STARTUP - fail closed, read once
standard-config ai/mcp-servers.json
container env Settings, MCP_* secrets
load_config validate, expand MCP_* urls
validate_credentials
McpRegistry, stores, PromptResolver, build_agent
OpenLIT prompt hub optional
HTTP
wires app
BAHMNI-AI FastAPI :8090 /bahmni-ai
4 POST /chat or /chat/confirm
5 SessionValidator session valid and gate privilege?
no
yes
401 session_expired
403 forbidden
502 openmrs_unavailable
6 ChatService: load + trim history
7 open MCP servers per run, per user
8 run agent instructions = PromptRenderer
9 model calls a tool?
no, answer ready
yes
next model round
10 tool visible? deny, overrides, allow
yes
no
11 validate arguments and constraints
denied or hidden error result to model
12 gate + tool privilege held?
yes
13 confirmation needed?
yes, not yet approved
no
14 fresh privilege check writes only
15 execute once timeout, never retried
16 audit log tool_audit result back to model
next model round
ApprovalRequired: store PendingConfirmation, TTL 300 s SSE confirmation_required
17 final text save conversation, tool results redacted
18 SSE done
OpenMRS /session re-validate
PostgreSQL ai_pending_confirmation
MCP servers tools/call
SSE stream: text, activity, tool_*, confirmation_required, done
spans, metrics
LLM provider Anthropic, OpenAI, Gemini, Ollama, compatible retry + fallback
PostgreSQL ai_conversation
error JSON
MCP servers connect, discover tools
PostgreSQL ai_conversation
cache miss
OpenLIT / OTLP optional, scrubbed
bahmni-core streams the reply back through bahmni-proxy
OpenMRS /session user, roles, privileges
Clinic user sees the answer word by word